

E-Polarized Plane-Wave Scattering by a Finite Sinusoidal PEC Grating via the Nyström Method

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Abstract—This paper presents a Nyström formulation for two-dimensional E-polarized plane-wave scattering by a finite sinusoidal grating made of a perfect electric conductor (PEC). The inverse-square-root endpoint singularity of the induced current is extracted as a factor, and the logarithmic integral equation is differentiated before the resulting Cauchy-singular equation is discretized at Gauss–Chebyshev nodes. The formulation accommodates arbitrary incidence angles. Convergence under order refinement and agreement with independently implemented analytical-regularization and pulse-basis methods are demonstrated for a flat strip, whose total scattering cross section approaches the geometrical-optics limit at high frequency. Results for a symmetric five-period grating show how corrugation height and electrical size redistribute the absolute far-field scattering pattern. Above the first propagating-harmonic threshold, broadened finite-grating lobes appear near the reference directions of the corresponding infinite grating, while finite-length effects preserve a central reflection lobe and an end-generated background. The study therefore clarifies how infinite-grating directions can be used to interpret scattering by short gratings.

Index Terms—Electromagnetic scattering, sinusoidal open strip, integral equations, Nyström method, perfect electric conductors.

I. INTRODUCTION

Finite corrugated conductors combine periodic-surface physics with end effects that are absent from an infinite grating. Even a scalar two-dimensional model therefore has to resolve a singular edge current, interaction across several corrugations, and radiation into both halfspaces. Early numerical treatments of two-dimensional diffraction and the associated quadrature techniques are surveyed by Nazarchuk [1]. Differentiated formulations based on the method of discrete singularities (MDS) and the Nyström method for open perfect-electric-conductor (PEC) arcs, including interpolation quadratures and supplementary conditions, were developed in [2]–[5]. Related edge-conditioned and exponentially convergent Nyström schemes for open structures are reported in [6]–[8]. Accurate regularized analyses of curved open PEC reflectors provide further full-wave benchmarks [9], [10].

The finite sinusoidal grating has also been studied with complementary methods. Kobayashi and Eizawa used a Wiener–Hopf perturbation analysis for a finite PEC sinusoid [11]; a later H-polarized analysis is reported in [12].

Vinogradova, Kobayashi, and Eizawa later developed a full-wave E-polarized solution by analytical regularization and examined the transition between finite- and infinite-grating behavior [13]. Related resonant configurations formed by two finite sinusoidal gratings are analyzed in [14]. These studies emphasize that periodic Floquet angles remain useful references, but a grating with only a few periods does not possess exact Floquet channels.

This paper gives a compact Nyström solution for one finite sinusoidal PEC grating. Its contributions are threefold. First, the geometry is symmetric, all size parameters are referred to the half-span L , and the incident angle is left arbitrary in the formulation. Second, accuracy is assessed by an L^2 density-convergence study and a three-backend comparison. Third, absolute polar patterns and the dimensionally correct total scattering cross section (TSCS) reveal separately the effects of corrugation height and electrical size for a five-period grating.

II. FORMULATION AND NYSTRÖM DISCRETIZATION

A. Geometry and Boundary-Value Problem

For a structure invariant along z , E polarization is represented by the scalar electric component $U = E_z$ in the x – y plane. The zero-thickness PEC arc is parameterized symmetrically as

$$\mathbf{r}(t) = (Lt, h \cos(\pi Pt)), \quad -1 \leq t \leq 1, \quad (1)$$

where $2L$ is its projected width, h/L is the relative corrugation height, and the integer P is the number of complete periods. Its projected period is $\Lambda = 2L/P$. Hence geometry is specified by h/L and P , while frequency enters only through the electrical size kL . The notation and angle convention are summarized in Fig. 1.

With the suppressed time factor $e^{-i\omega t}$, a unit-amplitude incident plane wave is

$$U^{\text{inc}}(\mathbf{r}) = \exp(ik\hat{\mathbf{d}} \cdot \mathbf{r}), \quad \hat{\mathbf{d}} = (\cos \beta, \sin \beta). \quad (2)$$

Thus constant-phase fronts travel along $\hat{\mathbf{d}}$. The reported computations use normal incidence, $\beta = \pi/2$, so the wave travels from the lower to the upper halfspace; no such restriction is made in the equations. The scattered field U^{sc} satisfies the Helmholtz equation, the outgoing Sommerfeld condition, local energy finiteness at both endpoints, and

$$U^{\text{inc}} + U^{\text{sc}} = 0 \quad \text{on } \Gamma. \quad (3)$$

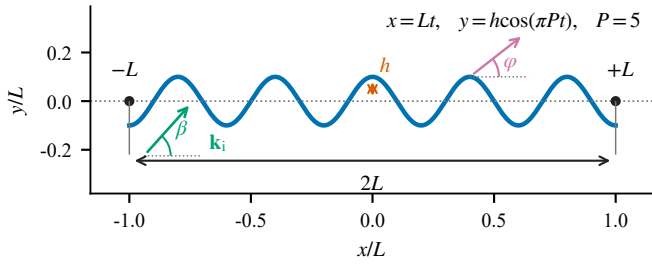


Fig. 1. Symmetric finite sinusoidal PEC grating and angle convention. The incident wave vector is $\mathbf{k}_i = k\hat{\mathbf{d}}$, where $\hat{\mathbf{d}} = (\cos \beta, \sin \beta)$; observation angle φ is measured counterclockwise from $+x$.

The outgoing Green kernel is consequently $H_0^{(1)}$.

B. Edge-Conditioned Integral Equation

Absorb the constant in the electromagnetic Green function into a scaled line density q . Its known endpoint behavior is factored as

$$q(t)|\mathbf{r}'(t)| = \frac{v(t)}{\pi\sqrt{1-t^2}}, \quad (4)$$

where v is smooth. Equation (4) extracts the singular factor; it does not remove the physical edge singularity. Define $R(t, \tau) = |\mathbf{r}(t) - \mathbf{r}(\tau)|$, $G(s) = H_0^{(1)}(ks)$, $w(t) = (1 - t^2)^{-1/2}$, and $u_0(\tau) = U^{\text{inc}}(\mathbf{r}(\tau))$. A single-layer representation and the PEC boundary condition give the logarithmically singular integral equation (IE)

$$\frac{1}{\pi} \int_{-1}^1 G(R(t, \tau)) v(t) w(t) dt = -u_0(\tau). \quad (5)$$

The explicit $1/\pi$ makes the continuous and discrete normalizations transparent.

Following the differentiated MDS and Nyström constructions for open PEC arcs in [2], [3], [5], the logarithmically singular IE (5) is differentiated with respect to τ , producing a Cauchy-singular IE. In particular, let

$$c_H = -\frac{2i}{\pi}, \quad (6)$$

$$K(t, \tau) = \frac{\partial}{\partial \tau} H_0^{(1)}(kR(t, \tau)) - \frac{c_H}{t - \tau}, \quad (7)$$

$$f(\tau) = -\frac{d}{d\tau} U^{\text{inc}}(\mathbf{r}(\tau)) = -ikU^{\text{inc}}(\mathbf{r}(\tau))\hat{\mathbf{d}} \cdot \mathbf{r}'(\tau). \quad (8)$$

The Cauchy-part extraction follows the normalized formulations in [2], [3]. With the Hankel normalization used here, the small-argument expansion of $H_0^{(1)}$ in [8] gives $c_H = -2i/\pi$; consequently, K has a finite diagonal limit. The differentiated equation is

$$\frac{1}{\pi} \text{p. v.} \int_{-1}^1 \left[\frac{c_H}{t - \tau} + K(t, \tau) \right] \frac{v(t)}{\sqrt{1-t^2}} dt = f(\tau). \quad (9)$$

Discretize the Cauchy-singular equation using the interpolation quadratures described in [2]–[4], with first-kind

Gauss–Chebyshev nodes and interlaced second-kind collocation points

$$t_i = \cos \frac{(2i-1)\pi}{2N}, \quad i = 1, \dots, N, \\ \tau_j = \cos \frac{j\pi}{N}, \quad j = 1, \dots, N-1. \quad (10)$$

Denote the density samples by $v_i = v(t_i)$. For $j = 1, \dots, N-1$, the boundary rows are

$$\frac{1}{N} \sum_{i=1}^N \left[\frac{c_H}{t_i - \tau_j} + K(t_i, \tau_j) \right] v_i = f(\tau_j). \quad (11)$$

Here the Gauss–Chebyshev weight π/N combines with the factor $1/\pi$ in (9), yielding the factor $1/N$.

As in [2], [3], a weighted supplementary condition restores the integration constant lost upon differentiation and hence equivalence to (5):

$$\frac{1}{N} \sum_{i=1}^N M(t_i) v_i = c, \quad (12)$$

where

$$M(t) = \int_{-1}^1 H_0^{(1)}(kR(t, \tau)) \frac{d\tau}{\sqrt{1-\tau^2}}, \quad (13)$$

$$c = - \int_{-1}^1 U^{\text{inc}}(\mathbf{r}(\tau)) \frac{d\tau}{\sqrt{1-\tau^2}}. \quad (14)$$

The logarithmic self term in M is subtracted analytically and its smooth remainder is evaluated by a higher-order Chebyshev rule. Equations (11) and (12) form an $N \times N$ dense system.

C. Far Field and TSCS

For $\hat{\mathbf{s}}(\varphi) = (\cos \varphi, \sin \varphi)$, define Φ_{sc} by $U^{\text{sc}}(r, \varphi) \sim e^{ikr} \Phi_{\text{sc}}(\varphi) / \sqrt{kr}$. The single-layer far-field asymptotics and the same quadrature in [3] give

$$\Phi_{\text{sc}}^{(N)}(\varphi) = \sqrt{\frac{2}{\pi i}} \frac{1}{N} \sum_{i=1}^N e^{-ik\hat{\mathbf{s}}(\varphi) \cdot \mathbf{r}(t_i)} v_i. \quad (15)$$

For unit incident amplitude, the total scattering cross section is

$$\sigma = \frac{1}{k} \int_0^{2\pi} |\Phi_{\text{sc}}(\varphi)|^2 d\varphi. \quad (16)$$

It has the required length dimension. The polar plots below display the associated absolute scattering diagram $10 \log_{10}[|\Phi_{\text{sc}}(\varphi)|^2/(kL)]$; they are not normalized to the maximum of each curve.

The verification, representative field pattern, height sweep, and electrical-size sweep are presented in Figs. 2–5, respectively.

For reference, the infinite periodic continuation has tangential harmonics

$$\xi_m = \cos \beta + \frac{m\pi P}{kL}, \quad |\xi_m| < 1. \quad (17)$$

Their upper- and lower-halfspace angles are $\arccos \xi_m$ and $2\pi - \arccos \xi_m$, respectively. These angles are only markers

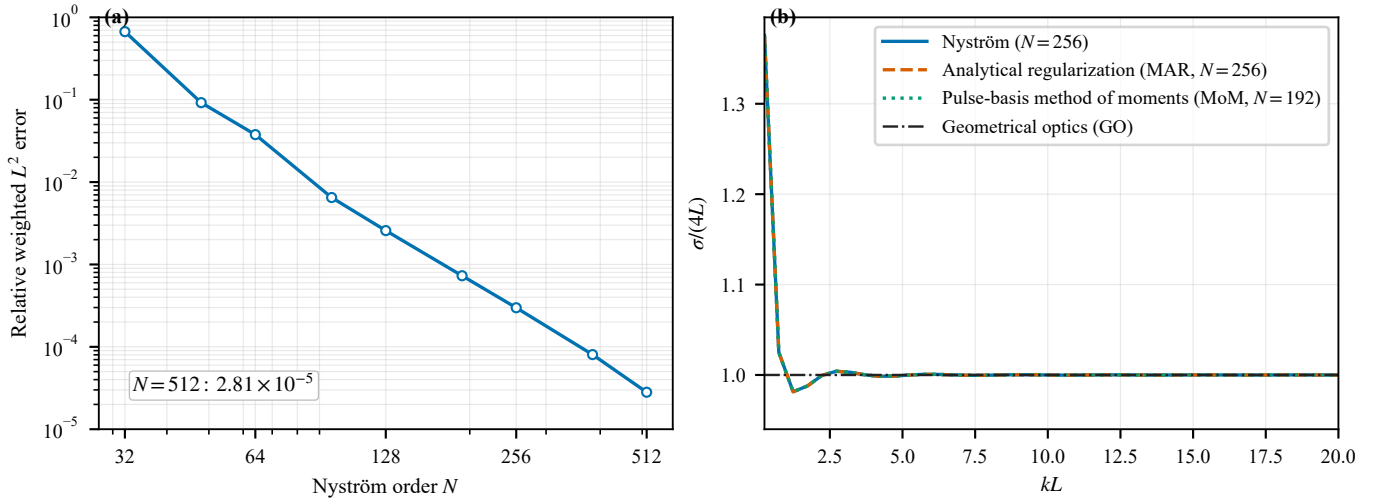


Fig. 2. Verification. (a) Relative L^2 error of the smooth density versus interpolation order for $P = 5$, $h/L = 0.10$, and $kL = 20$. (b) Flat-strip total scattering cross section (TSCS), normalized by its geometrical-optics limit $4L$, from the differentiated Nyström method ($N = 256$), the method of analytical regularization (MAR; $N = 256$), and a pulse-basis method of moments (MoM; 192 elements); the dash-dotted line is the geometrical-optics (GO) limit.

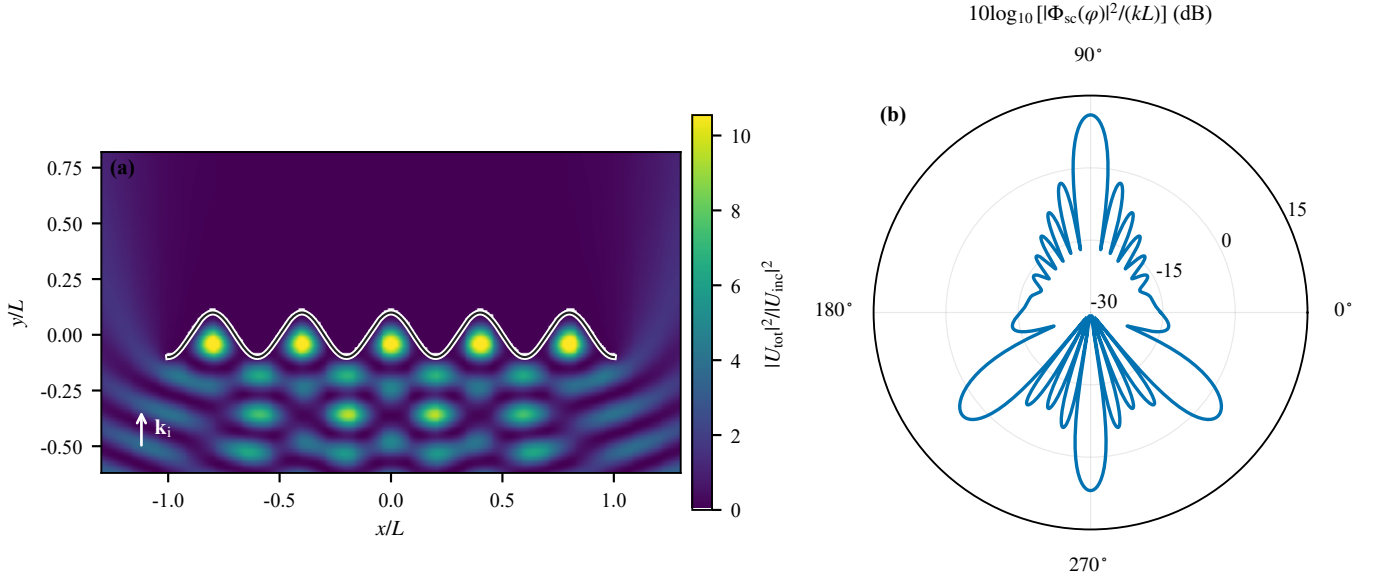


Fig. 3. Five-period grating at $P = 5$, $h/L = 0.10$, $kL = 20$, and $\beta = \pi/2$. (a) Relative total-field intensity $|U_{\text{inc}} + U_{\text{sc}}|^2/|U_{\text{inc}}|^2$. (b) Absolute polar scattering diagram based on Φ_{sc} in (15), shown as $10 \log_{10} [|\Phi_{\text{sc}}(\varphi)|^2/(kL)]$ on the common -30 to $+15$ dB scale.

for the broad lobes of the finite grating; below they are called infinite-grating reference directions. At normal incidence this reduces to

$$\cos \varphi_m = \frac{m\pi P}{kL}, \quad \left| \frac{m\pi P}{kL} \right| < 1. \quad (18)$$

The first-order markers become propagating at

$$kL = \pi P = 15.70796 \quad (P = 5). \quad (19)$$

III. NUMERICAL RESULTS

A. Convergence and Cross-Validation

Fig. 2(a) reports a relative weighted L^2 error of the smooth density for $P = 5$, $h/L = 0.10$, $kL = 20$, and $\beta = \pi/2$. Let $I_N v_N$ denote its Chebyshev interpolant and set $\theta_\ell = (\ell + 1/2)\pi/4096$. With the $N_{\text{ref}} = 800$ solution as the reference, the plotted error is

$$\epsilon_N = \left[\frac{\sum_{\ell=0}^{4095} |I_N v_N(\cos \theta_\ell) - I_{800} v_{800}(\cos \theta_\ell)|^2}{\sum_{\ell=0}^{4095} |I_{800} v_{800}(\cos \theta_\ell)|^2} \right]^{1/2}. \quad (20)$$

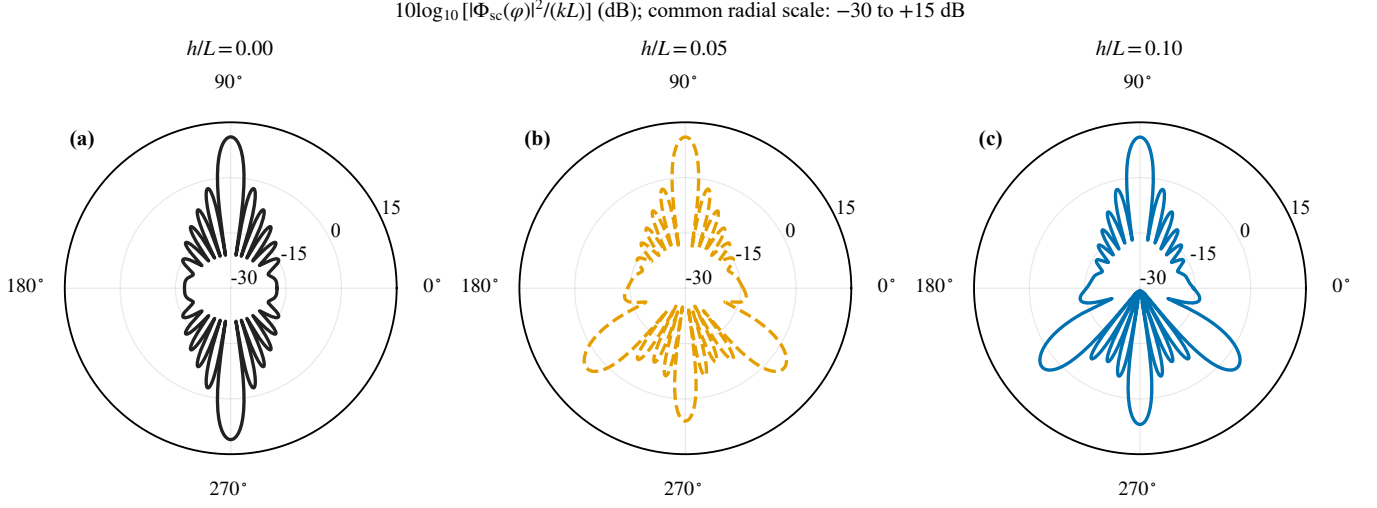


Fig. 4. Absolute polar scattering diagrams based on Φ_{sc} in (15), shown as $10 \log_{10}[|\Phi_{sc}(\varphi)|^2/(kL)]$, for $P = 5$, $kL = 20$, and $\beta = \pi/2$: (a) $h/L = 0$; (b) $h/L = 0.05$; and (c) $h/L = 0.10$. Each panel contains one curve and uses the common -30 to $+15$ dB scale.

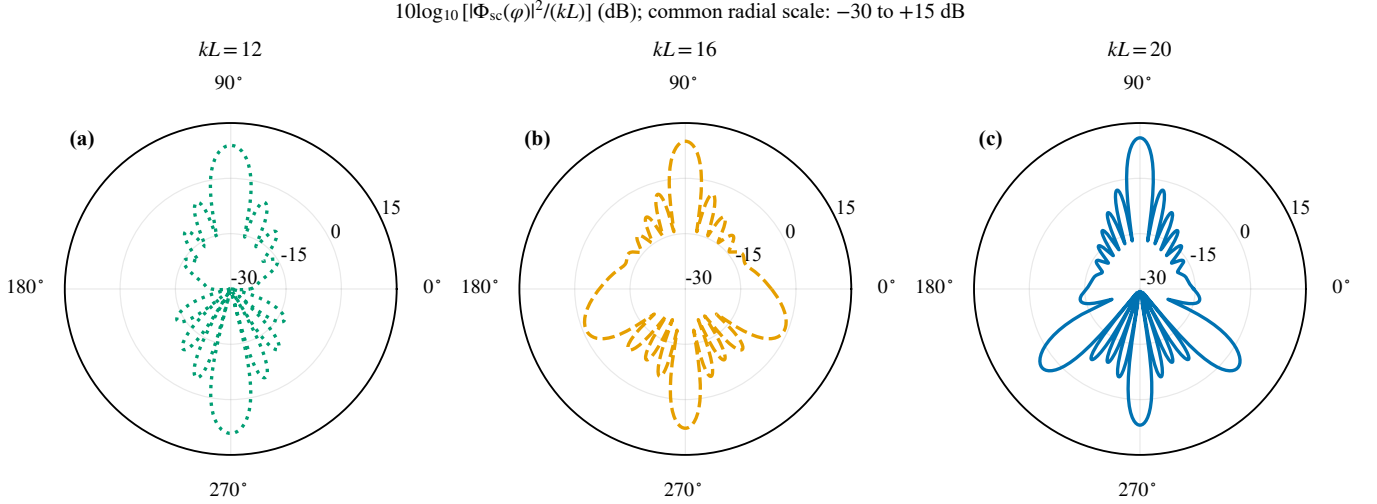


Fig. 5. Absolute polar scattering diagrams based on Φ_{sc} in (15), shown as $10 \log_{10}[|\Phi_{sc}(\varphi)|^2/(kL)]$, for $P = 5$, $h/L = 0.10$, and $\beta = \pi/2$: (a) $kL = 12$; (b) $kL = 16$; and (c) $kL = 20$. Each panel contains one curve and uses the common -30 to $+15$ dB scale. The first-order periodic-reference harmonics are cut off at $kL = 5\pi$ and propagate above it.

This midpoint sampling in θ approximates the $L^2_{1/\sqrt{1-t^2}}$ norm in t . For $N = 32, 48, 64, 96, 128, 192, 256, 384, 512$, the error decreases systematically and reaches 2.81142×10^{-5} at $N = 512$. The latter order is used for all corrugated-grating results below. Repeating every unique publication case at $N = 800$ changes the TSCS by at most 7.53315×10^{-6} and the full complex far-field pattern by at most 2.30664×10^{-5} in relative L^2 norm.

The second test uses a flat strip ($h = 0$) over 41 electrical sizes $0.25 \leq kL \leq 20$. Fig. 2(b) compares the present differentiated Nyström implementation with separately assembled analytical-regularization and pulse-basis method-of-moments backends. The maximum spread in $\sigma/(4L)$ among the three results is 1.58902×10^{-5} . The first two use $N = 256$, and the

pulse discretization uses 192 elements. At the most demanding endpoint, $kL = 20$, doubling those orders changes the TSCS by at most 5.03819×10^{-6} . At $kL = 20$ the Nyström value is 0.9999612. This is close to the geometrical-optics limit $\sigma/(4L) \rightarrow 1$ for a broadside-illuminated flat PEC strip of projected width $2L$ [8], but it is correctly retained as a finite-frequency value.

B. Representative Five-Period Grating

Fig. 3 shows the normal-incidence solution for $P = 5$, $h/L = 0.10$, and $kL = 20$. The total-field intensity is shown relative to the unit incident-field intensity. Below the grating, incident and reflected fields interfere; above it, the pattern contains the shadow lobe. The polar panel gives

TABLE I
NORMALIZED TSCS FOR THE HEIGHT AND ELECTRICAL-SIZE SWEEPS IN
FIGS. 4 AND 5 ($P = 5$, $\beta = \pi/2$).

$kL = 20$		$h/L = 0.10$	
h/L	$\sigma/(4L)$	kL	$\sigma/(4L)$
0	0.9999612	12	1.010974
0.05	0.9948396	16	0.9968797
0.10	0.9926829	20	0.9926829

the absolute scattering diagram based on Φ_{sc} in (15) over $0 \leq \varphi < 2\pi$. At this electrical size the $m = \pm 1$ periodic-reference harmonics are propagating, and the calculated finite-grating pattern contains broad off-normal structure near their infinite-grating reference directions. The central reflection lobe and end-generated background remain present because only five periods are retained.

C. Height and Electrical-Size Dependence

At fixed $P = 5$ and $kL = 20$, increasing h/L changes the phase and magnitude of the induced current across each corrugation. Fig. 4 shows the resulting redistribution among the specular and off-normal lobes. Because every panel has the same absolute radial scale, changes in level can be compared directly. The corresponding normalized TSCS values, together with those for the electrical-size sweep below, are listed in Table I.

Thus corrugation changes both the angular distribution and the integrated scattered power; the polar curves are not merely rescaled copies.

Frequency changes both the overall electrical width and the set of propagating periodic-reference harmonics. The three cases for $P = 5$ and $h/L = 0.10$ in Fig. 5 are also tabulated in Table I. At $kL = 12$ the first-order periodic harmonics are evanescent. They become propagating above the threshold in (19), so first-order finite-grating lobes appear in the $kL = 16$ pattern and separate further from their cutoff directions at $kL = 20$. Their finite width and interaction with the background preclude interpreting them as exact diffraction orders.

IV. CONCLUSION

An edge-conditioned integral equation, discretized with the Nyström technique, has been formulated for E-polarized plane-wave scattering by a finite sinusoidal PEC grating. The formulation accommodates arbitrary incidence, uses the outgoing $H_0^{(1)}$ kernel consistently with $e^{-i\omega t}$, and retains the correct $1/k$ factor in the TSCS. Density convergence, agreement

among three separately implemented formulations, and the approach of the flat-strip result to its geometrical-optics limit provide complementary accuracy checks. For the five-period grating, height mainly redistributes scattering among angular lobes, whereas electrical size determines when periodic-reference harmonics become propagating. Absolute polar plots and the reported $\sigma/(4L)$ values show that both redistribution and total scattered power vary with the profile.

REFERENCES

- [1] Z. T. Nazarchuk, *Numerical Investigation of Wave Diffraction from Cylindrical Structures*. Kyiv, Ukraine: Naukova Dumka, 1989, in Russian.
- [2] A. A. Nosich and Yu. V. Gandel, "Method of discrete singularities in the accurate modeling of a reflector beam waveguide," in *2005 IEEE MTT-S Int. Microwave Symp. Dig.*, Long Beach, CA, USA, 2005, pp. 1083–1086, doi: 10.1109/MWSYM.2005.1516860.
- [3] A. A. Nosich and Yu. V. Gandel, "Numerical analysis of quasioptical multireflector antennas in 2-D with the method of discrete singularities: E-wave case," *IEEE Trans. Antennas Propag.*, vol. 55, no. 2, pp. 399–406, Feb. 2007, doi: 10.1109/TAP.2006.889811.
- [4] A. A. Nosich, Yu. V. Gandel, T. Magath, and A. Altintas, "Numerical analysis and synthesis of 2D quasi-optical reflectors and beam waveguides based on an integral-equation approach with Nyström's discretization," *J. Opt. Soc. Amer. A*, vol. 24, no. 9, pp. 2831–2836, Sep. 2007, doi: 10.1364/JOSAA.24.002831.
- [5] Yu. V. Gandel, "Boundary-value problems for the Helmholtz equation and their discrete mathematical models," *J. Math. Sci.*, vol. 171, no. 1, pp. 74–88, 2010, doi: 10.1007/s10958-010-0127-3.
- [6] M. S. Tong and W. C. Chew, "Nyström method with edge condition for electromagnetic scattering by 2D open structures," *Prog. Electromagn. Res.*, vol. 62, pp. 49–68, 2006, doi: 10.2528/PIER06021901.
- [7] J. L. Tsalamengas, "Exponentially converging Nyström's methods for systems of singular integral equations with applications to open/closed strip- or slot-loaded 2-D structures," *IEEE Trans. Antennas Propag.*, vol. 54, no. 5, pp. 1549–1558, May 2006, doi: 10.1109/TAP.2006.874348.
- [8] O. V. Shapoval, R. Sauleau, and A. I. Nosich, "Scattering and absorption of waves by flat material strips analyzed using generalized boundary conditions and Nyström-type algorithm," *IEEE Trans. Antennas Propag.*, vol. 59, no. 9, pp. 3339–3346, Sep. 2011, doi: 10.1109/TAP.2011.2161547.
- [9] T. Oğuzer, A. Altıntaş, and A. I. Nosich, "Accurate simulation of reflector antennas by the complex source–dual series approach," *IEEE Trans. Antennas Propag.*, vol. 43, no. 8, pp. 793–801, Aug. 1995, doi: 10.1109/8.402198.
- [10] T. Oğuzer, A. I. Nosich, and A. Altıntaş, "E-polarized beam scattering by an open cylindrical PEC strip having an arbitrary "conical-section" profile," *Microw. Opt. Technol. Lett.*, vol. 31, no. 6, pp. 480–484, Dec. 2001, doi: 10.1002/mop.10067.
- [11] K. Kobayashi and T. Eizawa, "Plane wave diffraction by a finite sinusoidal grating," *IEICE Trans. Electron.*, vol. E74-C, no. 9, pp. 2815–2826, Sep. 1991, doi: 10.1587/e74-c_9_2815.
- [12] T. Eizawa and K. Kobayashi, "Wiener–Hopf analysis of the H-polarized plane wave diffraction by a finite sinusoidal grating," *Prog. Electromagn. Res.*, vol. 149, pp. 1–13, 2014, doi: 10.2528/PIER14063007.
- [13] E. D. Vinogradova, K. Kobayashi, and T. Eizawa, "Full wave analysis of plane wave diffraction by a finite sinusoidal grating: E-polarization case," *Wave Motion*, vol. 86, pp. 44–62, Mar. 2019, doi: 10.1016/j.wavemoti.2018.12.006.
- [14] E. D. Vinogradova and K. Kobayashi, "Complex eigenvalues of natural TM-oscillations in an open resonator formed by two sinusoidally corrugated metallic strips," *IET Microw., Antennas Propag.*, vol. 15, no. 10, pp. 1283–1298, 2021, doi: 10.1049/mia2.12166.